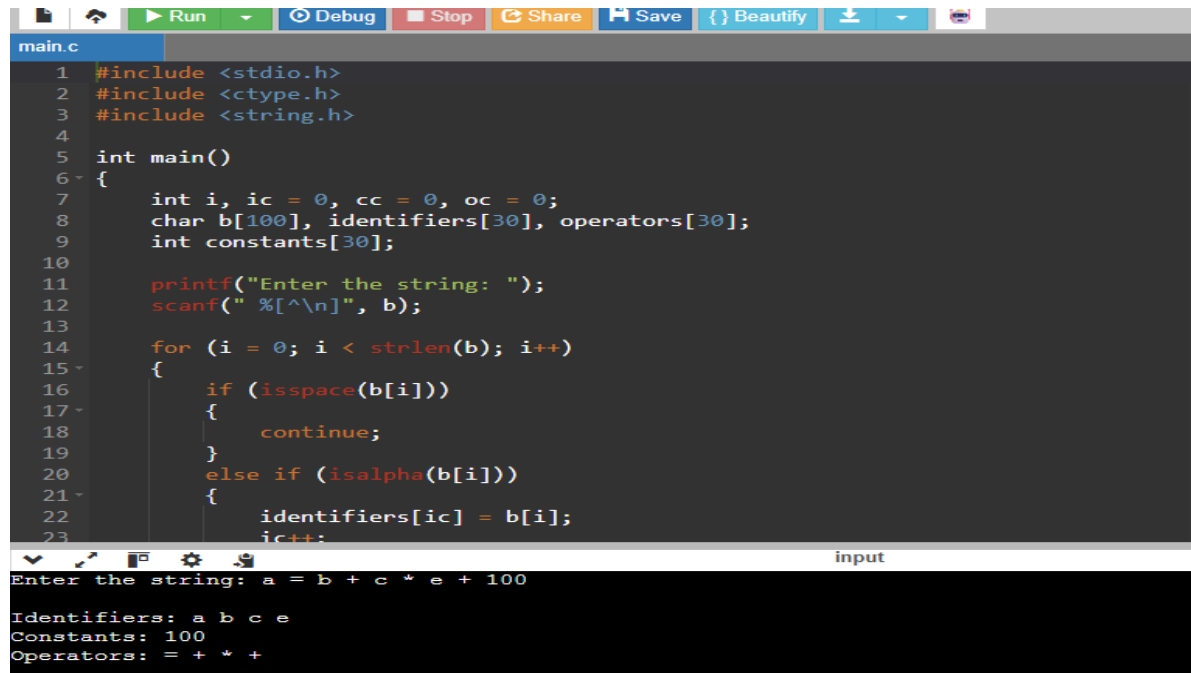


Experiment - 1:

Aim: To develop a lexical analyzer in C to identify identifiers, constants, and operators from a given input string.

Output:



```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4
5 int main()
6 {
7     int i, ic = 0, cc = 0, oc = 0;
8     char b[100], identifiers[30], operators[30];
9     int constants[30];
10
11     printf("Enter the string: ");
12     scanf("%[^\n]", b);
13
14     for (i = 0; i < strlen(b); i++)
15     {
16         if (isspace(b[i]))
17         {
18             continue;
19         }
20         else if (isalpha(b[i]))
21         {
22             identifiers[ic] = b[i];
23             ic++;
24         }
25         else if (isdigit(b[i]))
26         {
27             constants[cc] = b[i];
28             cc++;
29         }
30         else if (b[i] == '+' || b[i] == '-' || b[i] == '*' || b[i] == '/')
31         {
32             operators[oc] = b[i];
33             oc++;
34         }
35     }
36
37     printf("Identifiers: ");
38     for (i = 0; i < ic; i++) printf("%c ", identifiers[i]);
39     printf("\nConstants: ");
40     for (i = 0; i < cc; i++) printf("%c ", constants[i]);
41     printf("\nOperators: ");
42     for (i = 0; i < oc; i++) printf("%c ", operators[i]);
43 }
```

input

Enter the string: a = b + c * e + 100

Identifiers: a b c e

Constants: 100

Operators: = + * +

Experiment - 2:

Aim: To develop a lexical analyzer in C to identify whether a given line is a comment or not.

Output:



```
1 #include <stdio.h>
2 #include <string.h>
3
4 int main()
5 {
6     char com[100];
7     int i, found = 0;
8
9     printf("Enter comment: ");
10    fgets(com, sizeof(com), stdin);
11
12    if (com[0] == '/' && com[1] == '/')
13    {
14        printf("It is a comment");
15    }
16    else if (com[0] == '/' && com[1] == '*')
17    {
18        for (i = 2; i < strlen(com) - 1; i++)
19        {
20            if (com[i] == '*' && com[i + 1] == '/')
21            {
22                found = 1;
23                break;
24            }
25        }
26    }
27
28    if (found) printf("It is a comment");
29 }
```

input

Enter comment: /* This is a comment */

It is a comment

...Program finished with exit code 0